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PREFERENCES BY EGYPTIAN SPINY MICE FOR SOLUTIONS OF SUGARS, SALTS, AND ACIDS IN RICHTER-TYPE DRINKING TESTS

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Summary.—Egyptian spiny mice (*Acomys cahirinus*) were tested for drinking preferences by means of Richter-type tests (sapid solution versus distilled water). The subjects were 7 male and 7 female adults which were individually housed within an environmental chamber. Solutions used in the tests were concentrations of 5 common sugars (.005 M to 1.0 M), 3 salts (.005 M to 1.0 M), and 2 acids (4.0 pH to 1.5 pH). The molar range was too limited to identify the most preferred concentrations of fructose and glucose, but preferences for lactose, maltose, and sucrose appeared to occur at peak levels within the range. None of the salts (KCl, $MgSO_4$, and NaCl) was preferred over water in any of the tests, and both acids, the organic more than the inorganic, were rejected in all tests. The findings were contrasted with results from Richter-type tests conducted with other species of rodents.

This paper reports taste preferences of Egyptian spiny mice (*A. cahirinus*), as measured in Richter-type drinking tests (sapid fluids opposite distilled water). Egyptian spiny mice [also, Cairo spiny mice (Morris, 1965)], portrayed by Walker (1975) as the type species of the genus *Acomys*, belong to the subfamily *Murinae* (family *Muridae*) which is distributed principally on the continent of Africa (Ellerman, 1941; Piechocki, 1972). The few exceptions include *Rattus* and *Mus* from which the laboratorized variants of *R. norvegicus* and *M. musculus* have been derived. Like the other murines, *Acomys* possesses great adaptive capability. The four species of the genus inhabit a range which extends from northwest Pakistan through the Arabian Peninsula and northern Africa southward to Zimbabwe (Corbet, 1966; Walker, 1975). Within the range, *Acomys* occupies locales as diverse as sandy valleys, arid rocky sites, areas of dense vegetation, and human settlements (Shortridge, 1934; Walker, 1975). The group is prolific; the young are active at birth; and males are sexually mature at about seven weeks of age (Piechocki, 1972; Walker, 1975). Members of *Acomys* are alert, inquisitive, and docile (Piechocki, 1972) and resemble *Mus* in size and build (Shortridge, 1934). Scientific reference to spiny mice dates from Aristotle's mention (1910) of the bristly pelage which is a distinguishing feature of the genus and which may be described more exactly as a dorsal coat composed mainly of grooved spines (Rosevear, 1969). Owing to their reclusive and nocturnal mode of life, however, little is known about their behavior (Piechocki, 1972; Shortridge, 1934). Illustratively, assessments of food habits among spiny mice are tentative and are offered in terms which may

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reflect availability of food items rather than appetite preferences. Forms non-commensal with man have been observed feeding on different kinds of plants and animal matter, especially seeds, snails, and insects (Piechocki, 1972; Rosevear, 1969; Shortridge, 1934), and commensal forms have been said to feed ". . . on nearly everything they can find, including fibrous mats" (Walker, 1975, p. 935). Consequently, a study of taste preferences by need-free spiny mice may contribute information concerning the sensory characteristics of another member of the murine subfamily. Also, such investigation may help in an evaluation of palatibility factors in food selection by members of *Acomys.*

METHOD

The subjects were seven male and seven female Egyptian spiny mice which were between 150 and 180 days of age at the start of the preference tests. All of the animals were offspring of five excess pairs supplied from outbred stock kept at the Mohawk Park Zoo, Tulsa, Oklahoma. On the basis of the yellowish brown appearance of the dorsal coat, the subjects were identified as one of the commensal forms of *A. cahirinus* (Corbet, 1966). After weaning, the mice were reared in single-sex group cages until the start of the study. Thereafter, the animals were individually housed in adjoining stainless steel cages, measuring 18 cm × 18 cm × 38 cm long, within a Percival environmental chamber (3.0 m × 3.0 m × 6.1 m long) in which the ambient temperature was 24.5° C, and the relative humidity was 50.0%. The chamber was lighted from 0800 hr. to 2000 hr. daily. Food (Purina Laboratory Chow) and drinking fluids were available *ad libitum* to all subjects throughout the study. The fluids were provided in a pair of 140-ml calibrated Wahmann drinkers which were attached to the face of each cage. A clean pair of drinkers holding fresh fluids was presented every 48 hr. during both a period of pretesting and the preference tests which followed.

During 10 pretest days, all drinkers contained distilled water. At the start of the 11th day, all spiny mice were switched to a series of 52 consecutive 48-hr. Richter-type tests. After 24 hr. in each test, the drinkers were changed in position, and, at the end of the tests, fluid intakes, corrected for evaporative losses through use of control drinkers, were recorded. The sapid fluids used in the drinking preference tests were presented in an order determined by drawing lots. About 24 hr. before the start of a particular test, the selected chemical at the prescribed concentration was prepared by mixing the compound with distilled water. The solution was next either stored at room temperature or, in the case of the sugars, refrigerated and then returned to room temperature at the time of use. The salts and acids used in the tests were of reagent grade, and the sugars were of the highest purity commercially available. Chemicals used were as follows: (Sugars) D(−) fructose, D(+) glucose, D(+) lactose, D(+) maltose, and D(+) sucrose; (Salts) dried magnesium sulfate, potassium chloride, and sodium chloride; (Acids) citric acid and hydrochloric acid. In the course of testing, the sugars and salts were used at five different molar concentrations, and each acid was given at six different pH levels. Preference data used in statistical analyses were obtained by calculating sapid fluid intakes as percentages of total fluid intakes in the different tests. For each of the three categories of taste stimuli, the preference data were treated by means of an analysis of variance for a multifactor experiment having repeated measures on the same elements (Case II; Winer, 1971). Tests were also performed on simple effects and on trends.

RESULTS

In overview, the different analyses of variance showed in each case a significant Solutions × Concentrations interaction ($p < .01$), although all other interactions were insignificant ($p > .10$). Also, the sex difference main effect was insignificant in each analysis ($p > .10$).

Sugars

Inspection of Table 1 suggests that the range of molar concentrations used in the preference tests was wide enough to encompass the maximally preferred lactose, maltose, and sucrose solutions, but not those for glucose and fructose. Possibly, in the instances of glucose and fructose, concentrations above 1.0 M would have been required if the intention had been to locate the most acceptable solutions of these sugars. Analysis of the percentages of preference for sugars, summarized in Table 1, showed significant main effects ($df = 4/104, p < .01$) for sugars ($F = 28.07$) and concentrations ($F = 9.88$). Tests of simple effects were used to ascertain at which of the molar concentrations drinking

TABLE 1

MEAN PERCENTAGES OF PREFERENCE FOR FIVE SUGARS AND THREE SALTS AT FIVE MOLAR CONCENTRATIONS BY EGYPTIAN SPINY MICE GIVEN RICHTER-TYPE TESTS

Molarity	0.005 M		0.05 M		0 10 M		0.50 M		1 0 M	
	M%	*SE*	*M%*	*SE*	*M%*	*SE*	*M%*	*SE*	*M%*	*SE*
Sugars										
Fructose	52.4	6.4	83.2	5.4	88.9	3.9	93.5	1.1	93.0	2.4
Glucose	56.3	7.6	74.5	6.8	73.7	7.5	87.0	6.8	92.5	2.3
Lactose	32.6	8.0	43.8	8.5	80.9	5.1	75.7	5.2	57.3	4.2*
Maltose	75.0	5.3	83.6	4.3	92.0	2.0	92.1	3.1	72.8	5.3
Sucrose	66.5	7.9	85.1	5.6	92.6	2.2	91.7	2.0	77.0	7.4
Salts										
$MgSO_4$	42.5	7.2	15.5	5.1	19.3	6.6	7.8	1.6	5.7	1.6
KCl	36.9	7.0	39.6	7.3	24.0	4.4	15.6	4.3	25.8	7.3
NaCl	45.5	6.8	44.2	7.2	44.9	7.2	20.2	2.8	10.7	2.9

Note—% preference = test solution consumed (ml)/total fluid intake (ml) × 100.
*Saturated solution.

preferences differed significantly among sugars. At each of the concentrations in the range from .005 M through 0.50 M, there were insignificant differences in preferences among four of the five sugars (lactose was the exception) ($p > .10$). Lactose was significantly less preferred at each of the above concentrations and at 1.0 M as well ($p < .05$). At 1.0 M, the preference order among sugars was as follows: fructose = glucose > sucrose = maltose > lactose ($p < .05$). Table 1 also indicates that there were differences among sugars with respect to patterns of solution preference across, as well as within, concentrations. Analysis of trends showed significant linear and quadratic components for glucose and a significant quadratic component for maltose and for glucose ($p < .05$). Other trends were insignificant ($p > .10$).

Salts

Results from Richter-type tests with five different molar concentrations for each of the three salts are also condensed in Table 1. Analysis of the preferences for salts yielded a significant concentrations main effect ($F = 14.12$, $df = 4/312$, $p < .01$). The main effect for salts, however, was insignificant ($p > .10$). Tests of simple effects indicated significant differences in solution acceptance among the three salts at every concentration above 0.005 M. Further analysis of this finding through use of the Newman-Keuls procedure (Winer, 1971) showed that KCl and NaCl were accepted at rates which differed insignificantly ($p > .10$) in the range from 0.05 M through 0.50 M. Inspection of Table 1 suggests that there was an inverse relationship between solution acceptance and concentration strength for each of the three salts. The observation was supported by results from tests for trend in which only the linear component proved to be significant ($p < .05$).

Acids

Mean percentages of solution acceptance for citric acid and for HCl at six different pH levels are shown in Table 2. Analysis of data summarized in Table 2 showed that the main effect for concentrations was significant ($F = 12.35$, $df = 5/260$, $p < .01$) but that the main effect for acids was insignificant ($p > .10$). Tests of simple effects showed solution acceptance differed insignificantly between acids ($p > .10$) from 4.0 pH through 3.0 pH, but differed significantly ($p < .05$) from 2.5 pH through 1.5 pH. Tests of trend showed a significant linear component for citric acid and significant linear and quadratic components for HCl ($p < .05$). The other components were insignificant ($p > .10$).

TABLE 2

MEAN PERCENTAGES OF ACCEPTANCE FOR TWO ACIDS AT SIX LEVELS OF pH BY EGYPTIAN SPINY MICE GIVEN RICHTER-TYPE DRINKING TESTS

pH	Hydrochloric Acid		Citric Acid	
	M%	*SE*	*M%*	*SE*
4.0	44.5	9.1	47.6	7.9
3.5	50.6	4.7	46.0	7.6
3.0	35.9	7.4	23.2	5.0
2.5	47.8	6.9	17.7	5.0
2.0	34.2	7.7	8.9	1.7
1.5	13.3	4.3	2.6	1.1

Note.—% solution acceptance = test solution consumed (ml)/total fluid intake (ml) × 100.

Discussion

Generally, *Acomys* has been described, in both the noncommensal and commensal forms (Piechocki, 1972; Walker, 1975), as being opportunistic in its feeding habits. Even so, the present findings suggest that taste factors may influence food selection by spiny mice. Tests with the sugars elicited, in the main, positive appetitive responses, but the salts and acids triggered either indifferent or aversive appetitive reactions. Furthermore, there was indication that the spiny mice were able to make discriminative preferences among items within stimulus categories. Thus, although the sugars tended to be preferred, they were not equally acceptable at all five molar concentrations. More specifically, preferences for the two monosaccharides (fructose and glucose) were positively correlated with stimulus intensity. Preferences for each of the three disaccharides, however, could be plotted as a typical preference-aversion function along an inverted U-shaped curve (Jacobs, 1967). Consequently, at the strongest concentration level used in the study (1.0 M), all of the disaccharides were significantly less preferred than were either of the monosaccharides. The least preferred of the disaccharides was lactose which was drunk at significantly lower percentage rates at all five molar concentrations than was either of the two other disaccharides ($p > .05$). Even so, at the 0.10 M and 0.50 M concentrations, acceptance for solutions of lactose was more than twice the rate for water. In this connection, it may be noted that a 2:1 ratio of sapid solution intake over water in Richter-type tests defines a strong drinking preference (Engelmann, 1957). The test results for lactose conflict with results from comparable tests with laboratory rats for which latose is toxic (Jacobs, 1967). Unlike spiny mice, laboratory rats reject lactose solutions at all discriminable concentrations. Consequently, spiny mice may possibly be distinguished from laboratory rats, which are derived from the same murine subfamily, with respect to greater ability of the former to metabolize lactose.

Solution acceptance for salts and acids tended to be concentration-dependent. The stronger the concentration, the less acceptable were the chemicals in each of the two stimulus categories. Comparable results have been obtained for KCl and for $MgSO_4$ (Epsom salt) and for both citric acid and HCl in Richter-type tests with other species of rodents (Harriman, 1970, 1976, 1978; Harriman & Nevitt, 1977, 1978; Nevitt & Harriman, 1977). The diminished acceptance of KCl and of $MgSO_4$ with increase in molarity of concentration appears to be the result of both taste and postingestional factors functioning in some combination. While the role of taste is problematic in mediating intakes of the two salts, reduced drinking manifestly forestalled noxious effects from potassium excess (hyperkalemia) and from magnesium excess (hypermagnesemia). Also, like several other rodent species, the spiny mice rejected citric acid and HCl in increasingly strong fashion as pH was decreased. Similarily, the spiny mice showed greater aversion to the organic acid than to the inorganic acid at the same pH. The greater avoidance shown for citric acid in these comparisons probably is due to response to the total acid rather than to the free hydrogen ions primarily (Harriman, 1970).

The results from tests of solution acceptance for NaCl are not so readily reconciled with other reports of preferences for saline by rodents undergoing Richter-type tests. In an early study, Wolf and Lawrence (1963), summarizing the available literature on the subject, concluded that most species prefer hypotonic NaCl over water in two-bottle tests. Nonetheless, the spiny mice used in the present work were substantially indifferent to all three of the NaCl concentrations which fell within the hypotonic range. Moreover, the spiny mice, by drinking saline at less than half the rate of water, showed strong aversion (Engelmann, 1957) for both of the hypertonic solutions of NaCl. Contrary to the earlier opinion, these results may be aligned with an accumulating body of evidence which indicates differences among rodent species with respect to the acceptability of isomolar NaCl solutions. In different studies in which Richter-type tests were used, four cricetid species and one heteromyid species, like the spiny mice (a murine species), either rejected or were indifferent to hypotonic saline (Carpenter, 1956; Harriman, 1970, 1978; Harriman & Nevitt, 1977; Nevitt & Harriman, 1977). In contrast, Richter-type tests conducted with two other cricetid species (Harriman, 1976; Harriman & Nevitt, 1978) exhibited preferences for hypotonic saline resembling those shown by laboratory rodents belonging to the murine rodent group (Wolf & Lawrence, 1963). At present, neither preference nor aversions to hypotonic saline appear to be in any way systematically related to murid sub-family, tribe, or genus.

Indications from the present work that taste factors influence food selection by spiny mice may appear discrepant from field reports which have in the main classed *Acomys* as omnivores. In comment, it may be noted that the spiny mice used in the Richter-type tests were need-free animals insofar as could be arranged. Hence, under these conditions of laboratory maintenance, palatability may have contributed to the selection or rejection of the sapid fluids to a much greater extent than typically would be possible among feral spiny mice. More explicitly the field reports may have been based in some degree upon observations of spiny mice whose food choices were governed by gastric hunger rather than primarily by palatability factors. Nonetheless, palatability may affect selection of nutrients even among food-deprived spiny mice. There is evidence, as noted above, that feral *Acomys* fed preferentially on items within broad classes. Also, the low stimulus intensities required in the Richter-type tests to evoke preferences for several of the sugars, on one hand, and aversions to salts and to acids, on the other hand, may show selective pressure to use gustatory information for purposes of species survival. Preference rankings by spiny mice of items found ordinarily in the diets of commensal and non-commensal forms of *Acomys*, as by paired-comparison tests, should help verify whether or not taste preferences shown in Richter-type tests have explanatory significance for the food habits of the genus.

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